

WildSentinel: Safeguarding the Human–Wildlife Interface

Theme: Tech for Health and Humanity

Team: Code-Blooded

1. Problem Statement

Zoonotic diseases—those transmitted from animals to humans—represent a profound and escalating threat to global health and security. Epidemics such as COVID-19, SARS, and Ebola have exposed how human actions—especially **wildlife trade** and **habitat loss**—are dismantling the natural boundaries that once limited pathogen exchange. Expanding urbanization, illegal markets, and deforestation are forcing wildlife and people into closer contact, creating fertile ground for new infections to emerge.

In an increasingly connected world, these risks extend beyond ecology—they are a humanitarian concern. Yet current systems largely remain **reactive**, focusing on controlling outbreaks rather than preventing them. There is a critical need for **technological innovation** that uses data, analytics, and modeling to identify risks early and guide evidence-based interventions.

Our project, **WildSentinel: Safeguarding the Human–Wildlife Interface**, embodies the theme “*Tech for Health and Humanity*.” It transforms technology into a force for prevention, employing **AI-driven modeling, satellite imagery, and biosafety data integration** to predict zoonotic spillover hotspots before crises unfold. By turning complex environmental and trade data into actionable insights, **WildSentinel** advances human health while promoting coexistence with nature—protecting both people and the planet.

2. Objectives

The primary objective of **WildSentinel** is to **develop a predictive biosafety model** that quantifies and visualizes zoonotic spillover risks emerging from wildlife trade and habitat degradation.

Specific Objectives

1. **To identify and map global spillover hotspots** by integrating ecological, trade, and outbreak datasets.
2. **To quantify relationships** between trade intensity, deforestation, and zoonotic disease emergence.
3. **To simulate intervention scenarios**, such as restricting illegal trade or restoring forest cover, to estimate potential risk reduction.

4. **To produce biosafety recommendations** aligned with the *One Health* framework, uniting human, animal, and environmental health.
 5. **To design an open-source predictive tool** enabling policymakers and researchers to make data-driven decisions for early prevention.
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3. Proposed Methodology

The methodology integrates **data science, geospatial modeling, and biosafety analytics** into a cohesive workflow designed to predict and mitigate zoonotic spillover risks.

Data Integration

The project uses freely available global datasets:

- **CITES** – Wildlife trade and seizure statistics.
- **GBIF** – Species occurrence and biodiversity data.
- **NASA MODIS** – Satellite imagery for land-use and habitat loss trends.
- **WHO and PREDICT** – Records of past zoonotic outbreaks.
- **IUCN Red List** – Threatened species and ecosystem data.

These datasets will be standardized and merged into a single analytical framework using Python and QGIS.

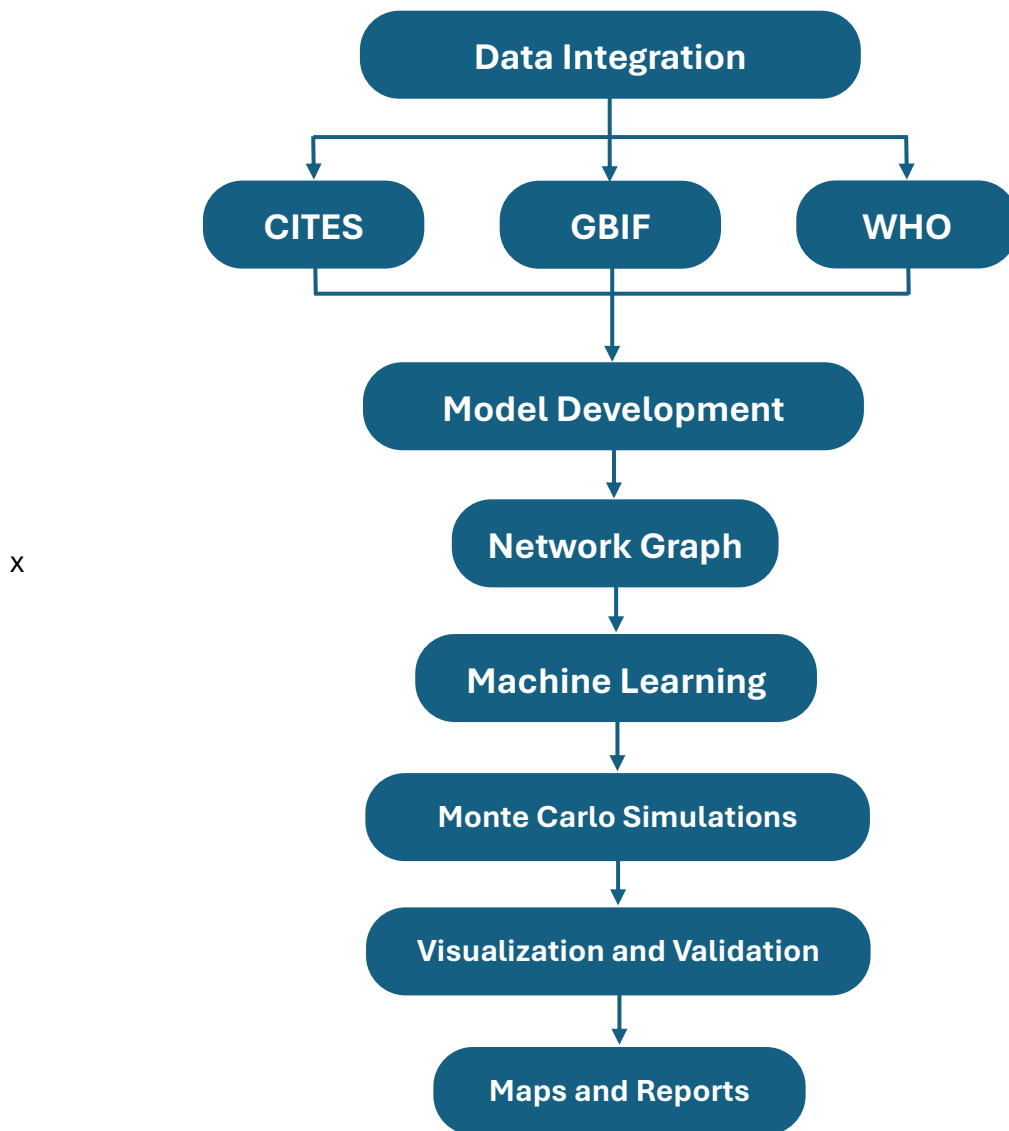
Model Development

- A **network graph** will represent interactions among wildlife species, markets, and human populations, using **NetworkX** to calculate connectivity and risk propagation.
- **Machine learning models** will predict spillover probabilities based on variables such as trade volume, forest fragmentation, and population density.
- **Monte Carlo simulations** will test “what-if” scenarios (e.g., 25% reduction in illegal trade) to quantify intervention impacts.

Visualization and Validation

- Hotspot and risk-density maps will be generated with **QGIS** for clear spatial visualization.
- Model performance will be validated using known outbreak data, aiming for over **75% predictive accuracy**.

- Ethical, transparent, and reproducible data practices will guide every stage of analysis.



4. Expected Results and Outcomes

Expected Results

1. **Global Risk Maps** highlighting regions where wildlife trade and habitat loss jointly elevate zoonotic spillover risks by 20% or more.
2. **Predictive Modeling Tool** capable of producing dynamic, real-time risk assessments using live satellite and trade data.
3. **Scenario Simulations** demonstrating potential reductions in spillover probability (20–25%) through targeted trade control and habitat restoration.

4. **Validation Reports** confirming the model's alignment with historical outbreak data and ensuring scientific reliability.

Expected Outcomes

- **Policy Impact:** Evidence-based insights supporting stronger regulations on wildlife trade and sustainable land-use planning.
 - **Scientific Contribution:** A reproducible, data-driven framework linking technology and biosafety for the *One Health* community.
 - **Public Health Advancement:** Tools for early warning and prevention, reducing economic and health burdens of future pandemics.
 - **Global Collaboration:** Open-source publication of data, models, and visual dashboards on platforms like GitHub to encourage shared solutions.
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Conclusion

WildSentinel stands as a technological sentinel—an intelligent guard at the fragile boundary between humans and wildlife. By merging data science with ecological insight, it shifts the global response from crisis management to **proactive prevention**.

This project exemplifies “*Tech for Health and Humanity*” by leveraging digital innovation to forecast and prevent outbreaks, protect vulnerable ecosystems, and secure a healthier, more sustainable coexistence between people and nature.